

LM Studio is large language model inference software for running and using large language models locally on personal computers. It provides tools for downloading and managing models, interacting with them through a chatbot interface, and serving locally running models to other applications through an application programming interface (API).

LM Studio uses llama.cpp as an inference engine for models in the GGUF format and also supports Apple's MLX framework on Apple silicon. In addition to its graphical interface, the platform includes command-line tools, software development kits for Python and TypeScript, and a local inference server with native, OpenAI-compatible, and Anthropic-compatible APIs.

History

LM Studio was launched in May 2023. It was initially developed as a desktop environment combining local model inference with model discovery, a chat interface, local model management, and an API server.

Support for the MLX inference framework on Apple silicon was introduced in October 2024, alongside its existing llama.cpp based inference system. Subsequent releases expanded the software's developer-oriented capabilities, including headless operation, on-demand model loading, command-line model downloading, tool calling, and speculative decoding.

In April 2026, Element Labs, the developer of LM Studio, acquired Locally AI, whose developer joined the company to work on native AI applications across Apple devices.

Features

LM Studio provides an integrated interface for finding and downloading AI models, loading models into memory, configuring inference settings, and interacting with models through a chat interface. Models can be obtained through integrations with repositories such as Hugging Face. The application can also use retrieval-augmented generation to provide local documents as context to language models.

The software can operate as a local inference server, allowing applications to access loaded models through LM Studio's REST API or compatibility interfaces modeled on the OpenAI and Anthropic APIs. Its Python and TypeScript software development kits provide programmatic model loading, text generation, embeddings, and agentic tool use.

LM Studio can also operate without its graphical interface. Its headless component, *llmster*, packages the core LM Studio runtime for use as a background service on local computers, servers, and GPU systems.

Related software and services

LM Studio has expanded beyond its original desktop application to include additional software and services. LM Link allows models running on another computer to be accessed remotely, while the company's mobile software grew from its acquisition of Locally AI in 2026.

The company also develops Bionic, an AI agent for document creation, computer programming, workflow automation, and computer interaction that can use locally running models through the LM Studio runtime.

Reception

PCWorld included LM Studio among applications for running language models locally, highlighting its model selection, configuration controls, developer interface, and support for retrieval-augmented generation, while describing it as oriented more toward advanced users than beginners.

In a comparison with Ollama, Windows Central highlighted LM Studio's graphical interface and its support for GPU acceleration on systems using integrated AMD and Intel graphics, describing it as a unified environment for discovering, installing, and interacting with local language models.

LM Studio has also been studied in research on local AI software. A 2026 digital-forensics

study compared LM Studio with Ollama and llama.cpp, examining artifacts left by the applications including model caches, configuration data, prompt histories, logs, and network activity.

See also

[SGLang](#)

[vLLM](#)

External links

[Official website](#)

[LM Studio documentation](#)

== References ==